5 CHEMICAL FORMULAE, EQUATIONS AND CALCULATIONS: PART 1

It is often important in chemistry to know how much of one substance reacts with a certain amount of another. To do this, we will learn about moles in this chapter.


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GOLD
GOLD

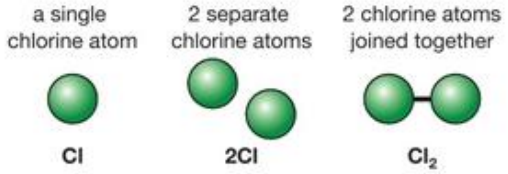
WHAT ALL THE NUMBERS MEAN

An example of a balanced chemical equation is:

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$$

When you write equations, it is important to be able to count how many of each type of atom you have. In particular, you must understand the difference between big numbers written in front of formulae (sometimes called **coefficients**), such as the **2** in **2**HCl, and the smaller, subscript (slightly lower on the line) numbers, such as the **3** in $CaCO_3$.

Another chemical equation is $2Cl \rightarrow Cl_2$. Here, **2Cl** represents 2 separate Cl atoms and **Cl_2** means that the atoms are joined together in a molecule.

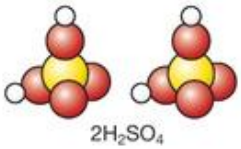


▲ Figure 5.2 The difference between 2Cl and Cl_2

Another balanced chemical equation is $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$.

Look at the way the numbers work in $2H_2SO_4$. The big number in front tells you that you have 2 sulfuric acid (H_2SO_4) molecules. The subscript 4 tells you that you have 4 oxygen atoms in each molecule. A small, subscript number in a formula applies only to the atom immediately before it in the formula. If you count the atoms in $2H_2SO_4$ you will find 4 hydrogens, 2 sulfurs and 8 oxygens.

If you have brackets in a formula, the small number refers to everything inside the brackets. For example, in the formula $Ca(OH)_2$, the 2 applies to both the oxygen and the hydrogen. The formula shows 1 calcium, 2 oxygens and 2 hydrogens. You will learn more about how to work out the formulae of compounds in Chapter 7.

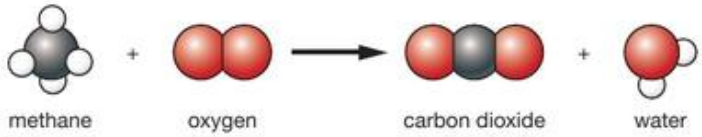


▲ Figure 5.3 $2H_2SO_4$. The sulfur atoms are shown in yellow, the oxygens in red and the hydrogens in white.

BALANCING EQUATIONS

Chemical reactions involve taking elements or compounds and moving their atoms around into new combinations. It follows that you must always end up with the same number of atoms that you started with.

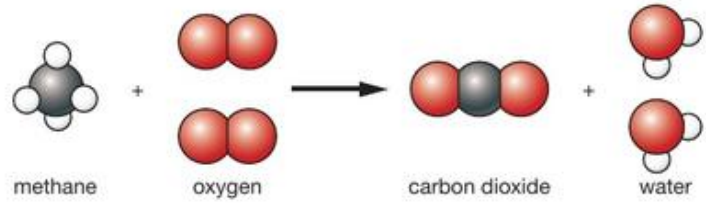
Imagine you had to write an equation for the reaction between methane, CH_4, and oxygen, O_2. Methane burns in oxygen to form carbon dioxide and water. Think of this in terms of rearranging the atoms in some models.



▲ Figure 5.4 Rearranging the atoms in methane and oxygen

If you count the atoms you had at the beginning (on the left-hand side of the arrow) and the atoms you have at the end (on the right-hand side of the arrow), you can see that this can't be right! During the rearrangement, we seem to have gained an oxygen atom and lost two hydrogens. The reaction must be more complicated than this. Since the substances are all correct, the proportions must be wrong.

Try again:



▲ Figure 5.5 A balanced equation for the reaction between methane and oxygen.

There are now the same number of each type of atom before and after. This is called **balancing the equation**.

In symbols, this equation would be:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

Think of each symbol (C or H or O) as representing one atom of that element. Count them up in the equation, and check that there is the same number of atoms on both sides.

HOW TO BALANCE EQUATIONS

In order to balance equations you should adopt a systematic approach.

- Work across the equation from left to right, checking one element after another, except if an element appears in several places in the equation. In that case, leave the element until the end and you will often find that it has sorted itself out.
- If you have a group of atoms (like a sulfate group (SO_4), for example), which is unchanged from one side of the equation to the other, there is no reason why you can't just count that up as a whole group, rather than counting individual sulfurs and oxygens. It saves time.
- Check everything at the end to make sure you haven't changed something that you have already counted.

HINT

This is really important! You must never, never change a formula when balancing an equation. All you are allowed to do is to write big numbers in front of the formula.

EXAMPLE 1

Balance the equation for the reaction between zinc and hydrochloric acid:

$$Zn + HCl \rightarrow ZnCl_2 + H_2$$

Work from left to right. Count the zinc atoms: 1 on each side; no problem!

Count the hydrogen atoms: 1 on the left, 2 on the right. If you have 2 at the end, you must have started with 2. The only way of achieving this is to have 2HCl. (You must not change the formula to H_2Cl because this substance does not exist.)

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2$$

Now count the chlorines: there are 2 on each side. Good! Finally check everything again to make sure and you've finished.

	$Zn + 2HCl$		$\rightarrow$	$ZnCl_2 + H_2$	
numbers of atoms	Zn	1		Zn	1
	H	2		H	2
	Cl	2		Cl	2

EXAMPLE 2

Balance the equation for the combustion of ethane:

$$C_2H_6 + O_2 \rightarrow CO_2 + H_2O$$

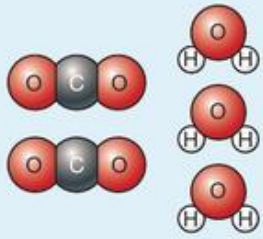
Starting from the left, balance the carbons:

$$C_2H_6 + O_2 \rightarrow 2CO_2 + H_2O$$

Now the hydrogens:

$$C_2H_6 + O_2 \rightarrow 2CO_2 + 3H_2O$$

Finally the oxygens: there are 7 oxygens ((2 × 2) + 3) on the right-hand side, but only 2 on the left. The problem is that the oxygens have to go around in pairs. So how can you obtain an odd number (7) of oxygens on the left-hand side?



$2CO_2$ $3H_2O$

▲ Figure 5.6 There are 7 O atoms in $2CO_2 + 3H_2O$

The trick with this is to allow yourself to have halves in your equation. 7 oxygen atoms, O, is the same as $3\frac{1}{2}$ oxygen molecules, O_2.

$$C_2H_6 + 3\tfrac{1}{2}O_2 \rightarrow 2CO_2 + 3H_2O$$

Now double everything to get rid of the half:

$$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$$

KEY POINT

In fact, it is acceptable to have halves in equations, but you don't usually come across them at Internaional GCSE.
You might reasonably argue that you can't have half an oxygen molecule, but to remove that problem all you have to do is double everything.

STATE SYMBOLS

HINT

Don't worry if this chemistry is new to you, or if at this stage you don't know what the state symbols should be. That is not important at the moment.

State symbols are often, but not always, written after the formulae of the various substances in an equation to show what physical state everything is in. You need to know four different state symbols:

(s) solid (l) liquid (g) gas (aq) in **aqueous** solution (dissolved in water)

So an equation might look like this:

$$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$$

This shows that *solid* potassium reacts with *liquid* water to make a *solution* of potassium hydroxide in water and hydrogen *gas*.

HINT

Remember that water is a liquid (l), not an aqueous solution (aq). An aqueous solution is formed when something is *dissolved in water*.

HOW MUCH OF EACH SUBSTANCE REACTS IN A CHEMICAL REACTION?

You can make iron(II) sulfide by heating a mixture of iron and sulfur:

$$Fe + S \rightarrow FeS$$

How do you know what proportions to mix them in? You can't just mix equal masses of them because iron and sulfur atoms don't weigh the same. Iron atoms contain more protons and neutrons than sulfur atoms, so an iron atom is one and three-quarter times heavier than a sulfur atom. In this or any other reaction, you can get the right proportions only if you know about the masses of the individual atoms that take part in the reaction.

RELATIVE ATOMIC MASS (A_r)

We have already looked at how to calculate relative atomic masses from the isotopic abundances in Chapter 3. Here we will look a little more closely at what exactly the relative atomic mass is.

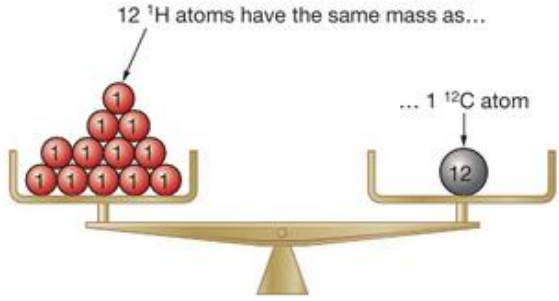
REMINDER

Remember that isotopes are atoms of the same element, but with different masses. Isotopes are explained in Chapter 3 (page 26).

EXTENSION WORK

This is a slight approximation. To be accurate, each of these hydrogen atoms has a mass of 1.008 on the carbon-12 scale. For International GCSE purposes, we take it as being exactly 1.

Atoms are amazingly small. The mass of a hydrogen atom is about 1.67×10^{-24} g (0.00000000000000000000000167 g). It is really difficult to use numbers such as this and so we use a scale of *relative* masses instead. The masses of atoms (and molecules) are compared with the mass of an atom of the carbon-12 isotope. We call this the carbon-12 scale. On this scale, one atom of the carbon-12 isotope weighs *exactly* 12 units and the mass of the most common hydrogen isotope is 1, which is a much simpler number to use!



▲ Figure 5.7 The most common hydrogen atom weighs one-twelfth as much as a ^{12}C atom.

The basic unit on the carbon-12 scale is $\frac{1}{12}$ of the mass of a ^{12}C atom, which is approximately the mass of the most common hydrogen atom. A fluorine-19 atom has a relative mass of 19 because its atoms have a mass 19 times that of $\frac{1}{12}$ of a ^{12}C atom. An atom of the most common isotope of magnesium weighs 24 times as much as $\frac{1}{12}$ of a ^{12}C atom, and is therefore said to have a relative mass of 24.

The masses we are talking about here are the masses of individual *isotopes*, but samples of an actual element contain different isotopes and so we need a measure of the average mass of an *atom* taking into account the different isotopes. This is the relative atomic mass.

The relative atomic mass of an element (as opposed to one of its isotopes) is given the symbol $\boldsymbol{A_r}$ and it is defined like this:

KEY POINT

Remember, to work out a weighted average we have to know how much of each isotope is present in a sample.

The relative atomic mass of an element is the weighted average mass of the isotopes of the element. It is measured on a scale on which a carbon-12 (^{12}C) atom has a mass of exactly 12.

Because we are talking here about *relative* masses they have no units.

On this scale, the relative atomic mass of chlorine is 35.45, that of lithium is 6.94 and that of sodium is 22.99 because we are taking into account the different isotopes, and are quoting an average mass for an atom. Although all elements consist of a mixture of isotopes, at International GCSE we only use relative atomic masses, including decimal places for Cl (35.5) and Cu (63.5), therefore we will take the relative atomic mass of lithium as 7 and that of sodium as 23.

HINT

Avoid the term 'relative molecular mass' because it can only properly be applied to substances which are actually molecules, that is, to covalent substances. You shouldn't use it for things like magnesium oxide or sodium chloride, which are ionic. Relative formula mass covers everything.

RELATIVE FORMULA MASS (M_r)

You can measure the masses of compounds on the same carbon-12 scale. For example, a water molecule, H_2O, has a mass of 18 on the carbon-12 scale. This means that a water molecule has 18 times the mass of $\frac{1}{12}$ of a ^{12}C atom.

If you are talking about compounds, you use the term *relative formula mass*. Relative formula mass is sometimes called relative molecular mass.

Relative formula mass is given the symbol $\boldsymbol{M_r}$.

CALCULATING SOME RELATIVE FORMULA MASSES

To find the relative formula mass (M_r) of magnesium carbonate, $MgCO_3$

Relative atomic masses: C = 12, O = 16, Mg = 24.

Add up the relative atomic masses to give the relative formula mass of the whole compound. In this case, you need to add up the masses of 1 × Mg, 1 × C and 3 × O.

$M_r = 24 + 12 + (3 \times 16)$
Mg C (3 × O)

$M_r = 84$

To find the relative formula mass of calcium hydroxide, $Ca(OH)_2$

Relative atomic masses: H = 1, O = 16, Ca = 40.

$M_r = 40 + (16 + 1) \times 2$
Ca O H

$M_r = 74$

To find the M_r of copper(II) sulfate crystals, $CuSO_4 \cdot 5H_2O$

This formula looks very different and a lot more complicated than the ones we have looked at above. When some substances crystallise from solution, water becomes chemically bound up with the salt. This is called **water of crystallisation**. The $5H_2O$ is water of crystallisation. The water is necessary to form crystals of copper(II) sulfate (and some other substances). There are always 5 water molecules associated with 1 $CuSO_4$ unit and they are part of the formula. Salts containing water of crystallisation are said to be **hydrated**. Other examples include $Na_2CO_3 \cdot 10H_2O$ and $MgCl_2 \cdot 6H_2O$.

Relative atomic masses: H = 1, O = 16, S = 32, Cu = 63.5.

It is easiest to work out the relative formula mass of water first:

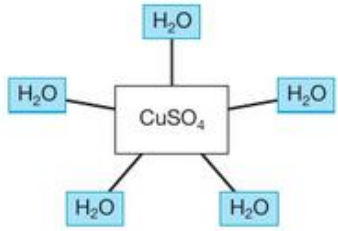
H_2O: $M_r = 2 \times 1 + 16 = 18$

Now add the correct number of waters on to the rest of the formula:

$CuSO_4 \cdot 5H_2O$: $M_r = 63.5 + 32 + (4 \times 16) + (5 \times 18) = 249.5$
Cu S (4 × O) (5 × H_2O)

HINT

Remember, you do not have to calculate the relative atomic masses, they will always be given to you in an exam, either in the question or on the Periodic Table. **If you use a Periodic Table, be sure to use the right number!** It will always be the larger of the two numbers given.



▲ Figure 5.8 A schematic diagram showing the formula of copper(II) sulfate crystals: there are five water molecules bonded to each copper(II) sulfate unit.

HINT

It is dangerous to calculate the number of hydrogens and oxygens separately. The common mistake is to work out 10 hydrogens (quite correctly!), but then only count 1 oxygen rather than 5.

USING RELATIVE FORMULA MASS TO FIND PERCENTAGE COMPOSITION

Having found the relative formula mass of a compound, we can work out the percentage by mass of any part of it. Examples make this clear.

To find the percentage by mass of copper in copper(II) oxide, CuO

Relative atomic masses: O = 16, Cu = 63.5.

M_r of CuO = 63.5 + 16 = 79.5

Of this, 63.5 is copper.

$$\text{Percentage of copper} = \frac{63.5}{79.5} \times 100$$

$$= 79.9\%$$

To find the percentage by mass of oxygen in sodium carbonate, Na_2CO_3

Relative atomic masses: Na = 23, C = 12, O = 16.

$M_r = (2 \times 23) + 12 + (3 \times 16)$

$= 106$

48 of this 106 is due to the oxygen (remember that there are 3 oxygens in the formula, so the total mass of oxygen is 3 × 16 = 48).

$$\text{Percentage of oxygen} = \frac{3 \times 16}{106} \times 100$$
$$= 45.3\%$$



▲ Figure 5.9 1-mole quantities (clockwise from upper left) of carbon (12 g), sulfur (32 g), iron (56 g), copper (63.5 g) and magnesium (24 g).

THE MOLE

In chemistry, the **mole** is a unit of the amount of substance. We can talk about an amount of substance in grams or an amount of substance in moles. The difference between expressing the amount of substance in moles or in grams is that 1 mole of any substance has its own particular mass. For example, 1 mole of water has a mass of 18 g, 1 mole of sulfur has a mass of 32 g and 1 mole of magnesium oxide has a mass of 40 g. The abbreviation for mole is **mol**.

We can therefore talk, for example, about an amount of water as 36 g or 2 mol, since 1 mol of water has a mass of 18 g. 36 g of sulfur, however, would only be 1.125 mol and 36 g of magnesium would be 1.5 mol because the mass of 1 mol of sulfur is 32 g and that of 1 mol of magnesium is 24 g.

Some people talk about the *amount in moles* and others talk about the *number of moles*. This is like talking about the *mass of a substance* or the *number of grams*. For example, we could say that the amount in grams of water is 36 g or the number of grams of water is 36 g. Similarly, we can say that the amount in moles of water is 2 mol or the number of moles of water is 2 mol.

CALCULATING THE MASSES OF A MOLE OF SUBSTANCE

You find the mass of 1 mole of a substance by calculating the relative formula mass (M_r) and attaching the units, grams.

1 mole of oxygen gas, O_2

Relative atomic mass: O = 16.

$$M_r \text{ of } O_2 = 2 \times 16$$
$$= 32$$

1 mole of oxygen, O_2, has a mass of 32 g.

1 mole of calcium chloride, $CaCl_2$

Relative atomic masses: Cl = 35.5, Ca = 40.

$$M_r \text{ of } CaCl_2 = 40 + (2 \times 35.5)$$
$$= 111$$

1 mole of calcium chloride has a mass of 111 g.

1 mole of iron(II) sulfate crystals, $FeSO_4 \cdot 7H_2O$

Relative atomic masses: H = 1, O = 16, S = 32, Fe = 56.

$$M_r \text{ of crystals} = 56 + 32 + (4 \times 16) + \{7 \times [(2 \times 1) + 16]\}$$

1 mole of iron(II) sulfate crystals has a mass of 278 g.

THE IMPORTANCE OF QUOTING THE FORMULA

Whenever you talk about a mole of something, you *must* quote its formula, otherwise there is a risk of confusion.

For example, if you talk about 1 mole of oxygen, this could mean:

1 mole of oxygen atoms, O, with a mass of 16 g

1 mole of oxygen molecules, O_2, with a mass of 32 g.

HINT

We often use the term 'molar mass' instead of 'mass of 1 mole'.

EXTENSION WORK

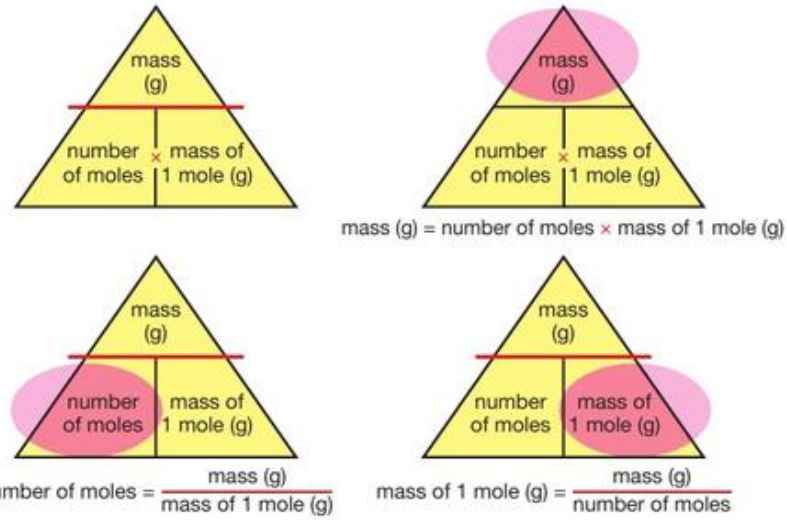
To be 100% correct 'molar mass' and 'mass of 1 mole' are not exactly the same thing and have different units: molar mass has units of g/mol and the mass of 1 mole has units of g. You are unlikely to come across the distinction between these at International GCSE.

SIMPLE CALCULATIONS WITH MOLES

You need to be able to interconvert between a mass in grams and an amount in moles for a given substance. There is a simple formula that you can learn:

$$\text{number of moles} = \frac{\text{mass (g)}}{\text{mass of 1 mole (g)}}$$

You can rearrange this formula to find whatever you want. If rearranging this expression causes you problems, you can learn a simple triangular arrangement to help you.



▲ Figure 5.10 Converting between mass in grams and number of moles.

Look at this carefully and make sure that you understand how you can use it to deduce the three equations that you might need.

Finding the mass of 0.2 mol of calcium carbonate, $CaCO_3$

Relative atomic masses: C = 12, O = 16, Ca = 40.

First find the relative formula mass of calcium carbonate.

$$M_r \text{ of } CaCO_3 = 40 + 12 + (3 \times 16)$$
$$= 100$$

1 mol of $CaCO_3$ has a mass of 100 g.

HINT

Don't get too reliant on learning formulae for doing simple calculations. If 1 mole weighs 100 g, it should be fairly obvious to you that 2 moles will weigh twice as much, 10 moles will weigh 10 times as much and 0.2 moles will weigh 0.2 times as much.

$$\text{Mass (g)} = \text{number of moles} \times \text{mass of 1 mole (g)}$$
$$= 0.2 \times 100$$
$$= 20\,\text{g}$$

0.2 mol of $CaCO_3$ has a mass of 20 g.

Finding the number of moles in 54 g of water, H_2O

Relative atomic masses: H = 1, O = 16.

1 mol of H_2O has a mass of 18 g.

HINT

If 1 mol weighs 18 g, then you need to find out how many times 18 goes into 54 to find out how many moles you have. On the other hand, if you feel safer using a formula, use it.

$$\text{Number of moles} = \frac{\text{mass (g)}}{\text{mass of 1 mole (g)}}$$
$$= \frac{54}{18}$$
$$= 3\,\text{mol}$$

54 g of water is 3 mol.

EXTENSION WORK

Moles and the Avogadro constant

Imagine you had 1 mole of ^{12}C. It would have a mass of 12 g and contain a very large number of carbon atoms, in fact about 6×10^{23} carbon atoms, which is 6 followed by 23 noughts. This number of atoms in 12 g of ^{12}C is called the **Avogadro constant**.

1 mole of anything else contains this same number of particles. For example:

1 mole of magnesium contains 6×10^{23} magnesium atoms, Mg, and has a mass of 24 g.

1 mole of water contains 6×10^{23} water molecules, H_2O, and has a mass of 18 g.

This is the reason that we use moles: if we know the number of moles we also know how many particles are present. If we know how many particles are present, we can work out how much of one substance reacts with a certain amount of another.



▲ Figure 5.11 Approximately 1×10^{32} water molecules go over Niagara Falls every second during the summer. That's 100 million million million million million water molecules per second.

FORMULAE

The formula of sulfur dioxide is SO_2. We can see that there are 2 O atoms for each S atom. If we had 1 mol SO_2 we would still have twice as many O atoms as S atoms; there would be 1 mol S atoms and 2 mol O atoms. If we know that in a certain sample of sulfur dioxide we have 0.1 mol of S atoms we also know that we must have 0.2 mol O atoms. 0.2 mol contains twice as many atoms as 0.1 mol.

If we did an experiment and found that a compound contained 0.2 mol Ca and 0.2 mol O, then we could work out that the formula must be CaO because there are the same number of Ca atoms as O atoms.

If a compound contains 0.4 mol Mn and 0.8 mol O then the formula must be MnO_2 because there are twice as many O atoms as Mn atoms.

The formulae that we have found here are called *empirical formulae*.

The empirical formula shows the simplest whole number ratio of the atoms present in a compound.

There is another type of formula that we will come across, called the *molecular formula*.

The molecular formula shows the actual number of atoms of each element present in a molecule (covalent compound) or formula unit (ionic compound) of a compound.

The molecular formula can be the same as the empirical formula or a multiple of the empirical formula. The empirical formula of calcium oxide is CaO and the molecular formula is also CaO. The empirical formula of hydrogen peroxide is HO and the molecular formula is H_2O_2.

In order to work out the molecular formula from the empirical formula we need more information – the M_r of the compound. We will look at this again below.

WORKING OUT EMPIRICAL FORMULAE

In order to find out the empirical formula of a compound such as copper oxide, we need to know how many atoms of copper combine with how many atoms of oxygen. We can work out the number of atoms from the number of moles. If we know the ratio between the number of copper atoms and oxygen atoms in the compound we know the formula.

KEY POINT

0.02 mol of atoms is twice as many atoms as 0.01 mol of atoms.

HINT

More significant figures should have been used for the number of moles in the examples on this page but these have been omitted for simplicity.

EXAMPLE 3

A sample of a compound contains 1.27 g of Cu and 0.16 g of O. Calculate the empirical formula. (A_r of Cu = 63.5, A_r of O = 16)

It is easiest to do this in columns using a table:

	Cu	O
masses/g	1.27	0.16
find the number of moles of atoms by dividing the mass by the mass of 1 mole of atoms	1.27/63.5	0.16/16
number of moles of atoms	0.02	0.01
divide by the smaller number to find the ratio	0.02/0.01	0.01/0.01
ratio of moles	2	1
empirical formula	Cu_2O	

From calculating the number of moles of Cu and O we can see that there are twice as many moles of Cu and therefore there must be twice as many Cu atoms as O atoms in the compound.

It is important to remember in this calculation that we are working out how many *atoms* of copper combine with how many *atoms* of oxygen so we divide the mass of oxygen by 16 (the mass of 1 mole of oxygen atoms) rather than by 32 (the mass of 1 mole of oxygen molecules).

EXAMPLE 4

A sample of a compound contains 0.78 g of K, 1.10 g of Mn and 1.28 g of O. (A_r of K = 39, A_r of Mn = 55, A_r of O = 16)

Again, we use a table:

	K	Mn	O
masses/g	0.78	1.10	1.28
find the number of moles of atoms by dividing the mass by the mass of 1 mole of atoms	0.78/39	1.10/55	1.28/16
number of moles of atoms	0.02	0.02	0.08
divide by the smallest number to find the ratio	0.02/0.02	0.02/0.02	0.08/0.02
ratio of moles	1	1	4
empirical formula	$KMnO_4$		

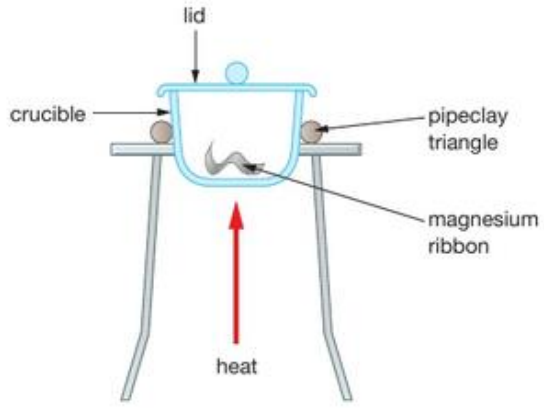
The empirical formula of this compound is $KMnO_4$.

ACTIVITY 3

▼ PRACTICAL: INVESTIGATING THE FORMULA OF A METAL OXIDE BY COMBUSTION

!

Safety Note: Wear eye protection and take care not to get burned when raising the crucible lid with tongs. The crucible will stay very hot for some time.



▲ Figure 5.12 Apparatus for determining the formula of magnesium oxide

KEY POINT

How do we know when the reaction has finished? We can generally see when there is no more reaction because nothing is happening in the crucible. We can also check by weighing the crucible and contents, then heating it and weighing again. If the mass is higher when we weigh it again then this indicates that a reaction has occurred. We keep heating and weighing until the mass stays constant.

We can find the formula of magnesium oxide by burning magnesium in oxygen and looking at how the mass changes. The following procedure is usually used:

- Weigh a crucible with a lid.
- Place a piece of magnesium ribbon about 10 cm long in the crucible and weigh the crucible and contents.
- Set up the apparatus as shown in Figure 5.12.
- Heat the crucible strongly (a roaring flame).
- Lift the lid every few seconds.
- When the reaction is finished, allow the crucible and contents to cool.
- Weigh the crucible and contents.

When the magnesium burns it does so with a bright white flame. Magnesium oxide, a white powder, is produced in the reaction.

A lid is placed on the crucible to prevent the white powder (magnesium oxide) escaping but the lid must be lifted every few seconds to allow oxygen into the crucible to react with the magnesium.

A set of results for this practical could be:

mass of empty crucible/g	32.46
mass of crucible + magnesium/g	32.70
mass of crucible + contents at end of experiment/g	32.86

We can work out the mass of magnesium by subtracting the mass of the crucible from the mass of the crucible + magnesium:

mass of magnesium = 32.70 – 32.46 = 0.24 g

The reason that the mass increases is because the magnesium combines with the oxygen in the air. The mass of magnesium oxide is greater than the mass of just magnesium due to the extra oxygen.

KEY POINT

Remember: Number of moles is mass in grams divided by the mass of 1 mole in grams.

The overall reaction that occurred was:

$$2Mg + O_2 \rightarrow 2MgO$$

magnesium + oxygen → magnesium oxide

The type of reaction that occurred was combustion.

When we do this experiment in practice, it is not very often that the ratio comes out as 1:1. There are things that may go wrong with this experiment which could affect the results, such as:

- some of the magnesium oxide powder might escape when the lid is lifted
- not all the magnesium might have reacted
- the magnesium can also react with nitrogen in the air.

HINT

The original masses were measured to 2 significant figures so the number of moles should also be given to 2 significant figures (0.010). If you write 0.01 for the number of moles of atoms in the exam, this will be fine and still give you the correct answer.

!

Safety Note: The teacher demonstrating needs to wear a face shield and use safety screens. The pupils require eye protection and should be no closer than 2 metres. If a drying agent is needed anhydrous calcium chloride should be used NOT concentrated sulfuric acid.

KEY POINT

We must always make sure that the tube is filled with hydrogen before lighting the stream of hydrogen because a mixture of hydrogen and oxygen is explosive! It is therefore important to let the stream of hydrogen gas flow through the tube for a little while (to flush out all the oxygen) before lighting it or lighting the Bunsen burner.

We can work out the mass of oxygen that combines with the magnesium:

mass of oxygen = 32.86 – 32.70 = 0.16 g

In order to find the formula of magnesium oxide we need to work out how many moles of magnesium atoms combine with how many moles of oxygen atoms.

	Mg	O
masses/g	0.24	0.16
find the number of moles of atoms by dividing the mass by the mass of 1 mole	0.24/24	0.16/16
number of moles of atoms	0.010	0.010
divide by the smaller number to find the ratio	0.010/0.010	0.010/0.010
ratio of moles	1	1
empirical formula	MgO	

The empirical formula of magnesium oxide is MgO.

THE FORMULA FOR COPPER OXIDE

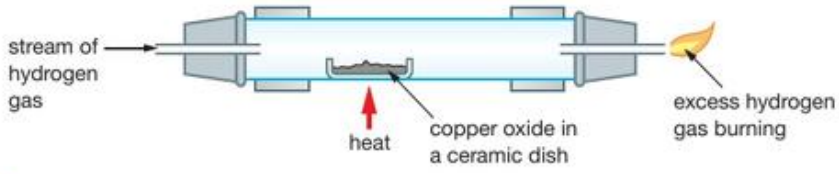
ACTIVITY 4

▼ PRACTICAL: INVESTIGATING THE FORMULA OF A METAL OXIDE BY REDUCTION

We can also find the formula of an oxide by removing the oxygen from it and looking at how the mass changes.

The following procedure can be used to find the formula of copper oxide:

- Weigh a ceramic dish.
- Put about 3 g of copper oxide in the ceramic dish and weigh the dish again.
- Place the ceramic dish in a tube as shown in Figure 5.13.
- Pass hydrogen gas over the copper oxide.
- Ignite the excess hydrogen, which comes out of the small hole in the boiling tube.
- Heat the copper oxide strongly until the reaction is finished (pink-brown copper metal will be seen).



▲ Figure 5.13 The experimental set-up for finding the formula of copper oxide.

A set of results for this practical could be:

mass of empty dish/g	23.78
mass of dish + copper oxide/g	26.96
mass of dish + contents at end of experiment/g	26.32

KEY POINT

This is a displacement (or competition) reaction. Hydrogen is more reactive than copper and displaces it from copper oxide. Displacement reactions are discussed in Chapter 14.

The reason that the mass decreases is because the hydrogen combines with the oxygen from the copper oxide to form water. The oxygen is removed from the copper oxide and we are left with only copper in the dish at the end of the experiment. Because oxygen is removed from the copper oxide we say that the copper oxide has been reduced. Reduction is explained in Chapter 14.

From this data we can calculate the mass of copper oxide at the beginning by subtracting the mass of the dish from the mass of the dish + copper oxide:

mass of copper oxide = 26.96 – 23.78 = 3.18 g

We can work out the mass of copper remaining at the end by subtracting the mass of the dish from the mass of the dish + copper at the end.

mass of copper = 26.32 – 23.78 = 2.54 g

The mass has decreased because the oxygen has been removed from the copper oxide and we can calculate the mass of oxygen by subtracting the mass of copper at the end from the mass of copper oxide:

mass of oxygen = 3.18 – 2.54 = 0.64 g

We now know that 2.54 g of copper combines with 0.64 g of oxygen in copper oxide and can deduce the empirical formula:

	Cu	O
masses/g	2.54	0.64
find the number of moles of atoms by dividing the mass by the mass of 1 mole	2.54/63.5	0.64/16
number of moles of atoms	0.0400	0.040
divide by the smaller number to find the ratio	0.0400/0.040	0.040/0.040
ratio of moles	1	1
empirical formula	CuO	

KEY POINT

The equation for the reaction that occurs in this experiment is:

$$CuO + H_2 \rightarrow Cu + H_2O$$

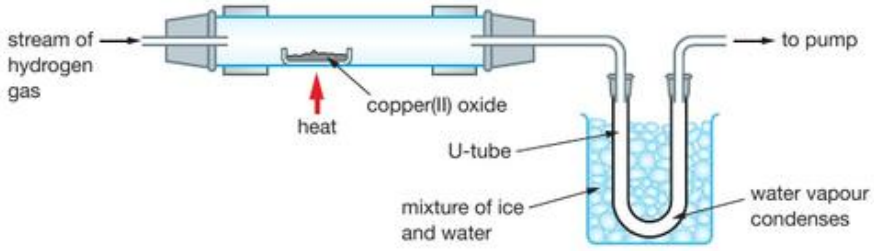
copper(II) oxide + hydrogen → copper + water

As well as a displacement reaction, this type of reaction can also be called a redox reaction. Redox reactions are discussed in Chapter 14.

Safety Note: The teacher demonstrating needs to wear a face shield and use safety screens. The pupils require eye protection and should be no closer than 2 metres. If a drying agent is needed anhydrous calcium chloride should be used NOT concentrated sulfuric acid.

DETERMINING THE FORMULA OF WATER

We can modify the apparatus in Figure 5.13 to allow us to determine the formula of water.



▲ Figure 5.14 Apparatus that can be used to determine the formula of water.

The experiment is conducted in the same way except that this time the water vapour that is produced from the reaction between copper(II) oxide and hydrogen is condensed. The contents of the dish at the beginning and the end of the experiment are again weighed but this time the mass of water that collects in the U-tube must also be measured.

mass of empty dish/g	23.78
mass of dish + copper oxide/g	26.96
mass of dish + contents at end of experiment/g	26.32
mass of water/g	0.72

Here we are using the same results as above. The mass of oxygen lost from the copper(II) oxide is 0.64g.

All the oxygen lost from the copper(II) oxide combines with hydrogen to form water. This means that the water contains 0.64g of oxygen. 0.72g of water was collected so the mass of hydrogen in the water must be 0.72 – 0.64 = 0.08g. We can now determine the empirical formula of water:

	H	O
masses/g	0.08	0.64
number of moles of atoms	0.08/1	0.64/16
number of moles of atoms	0.08	0.040
divide by the smaller number to find the ratio	0.08/0.040	0.040/0.040
ratio of moles	2	1
empirical formula	H_2O	

WORKING OUT FORMULAE USING PERCENTAGE COMPOSITION FIGURES

In the worked examples and practical examples above, we have determined the empirical formulae of compounds using masses. However, we are often given percentages by mass instead of just masses.

EXAMPLE 5

Find the empirical formula of a compound containing 82.7% C and 17.3% H by mass (A_r of H = 1, A_r of C = 12).

The percentage figures apply to any amount of substance you choose, so let's choose 100 g. In this case the percentages convert simply into masses: 82.7% of 100 g is 82.7 g (Table 5.1).

Table 5.1 Finding the ratio from percentage by mass

	C	H
percentages/%	82.7	17.3
masses in 100 g/g	82.7	17.3
number of moles of atoms	82.7/12	17.3/1
number of moles	6.89	17.3
divide by smallest to get ratio	6.89/6.89	17.3/6.89
ratio of moles	1	2.5

From this we can see that there are 2.5 mol of H atoms for every mole of C atoms. The empirical formula, however, is the *whole number ratio* of the elements present in a compound. To obtain a whole number here we multiply both numbers by 2 to get C_2H_5, which is the empirical formula.

HINT

When you calculate 17.3/6.89 you actually obtain 2.51. When doing these calculations it is fine to round 2.51 to 2.5 or 1.01 to 1 but what you must not do is round 2.51 to 3!
If you got a ratio of C:H of 1:1.33 you would not round this down to 1:1 but rather multiply by 3 to obtain C_3H_4.

CONVERTING EMPIRICAL FORMULAE INTO MOLECULAR FORMULAE

When you have learned a bit more organic chemistry in Unit 4 you will realise that, in the example we have just looked at, C_2H_5 can't possibly be the real formula of the compound. The molecular formula (the actual number of each atom present in a molecule) has to be a multiple of C_2H_5, like C_4H_{10}.

We can find the molecular formula if we know the relative formula mass of the compound (or the mass of 1 mole, which is just the M_r in grams).

In the previous example, suppose you were told that the relative formula mass of the compound was 58.

C_2H_5 has a relative formula mass of 29 (A_r of H = 1, A_r of C = 12).

All you need to find out is how many times 29 goes into 58.

$$58/29 = 2$$

So you need 2 lots of C_2H_5, in other words, C_4H_{10}.

The molecular formula is C_4H_{10}.

EXAMPLE 6

A compound has the empirical formula CH_2. If the relative formula mass is 56, work out the molecular formula.

The relative formula mass of CH_2 is $12 + (2 \times 1) = 14$.

$$56/14 = 4$$

Therefore there must be 4 lots of CH_2 in the actual molecule and the molecular formula is C_4H_8.

Remember, the ratio of elements in the molecular formula must be the same as in the empirical formula: C_4H_8 cancels down to CH_2.

EMPIRICAL FORMULA CALCULATIONS INVOLVING WATER OF CRYSTALLISATION

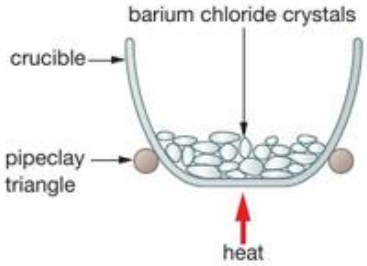
FINDING THE *n* IN $BaCl_2 \cdot nH_2O$

REMINDER

Remember that when some substances crystallise from solution, water becomes chemically bound up with the salt. This is called **water of crystallisation**. The salt is said to be **hydrated**.

REMINDER

Anhydrous means without water.



▲ Figure 5.15 Heating barium chloride crystals in a crucible

!

Safety Note: Barium chloride is toxic but a Bunsen flame is not hot enough to release it into the atmosphere.

HINT

If you are doing this experiment, how can you be sure that all the water has been driven off? The best way to do this is to heat it to constant mass. You heat the crucible and weigh it. Keep doing this until the mass remains constant. If the mass has gone down after heating there may still be water present, but when the mass remains constant after heating you know that all the water has been driven off.

Usually, when you heat a salt that contains water of crystallisation, the water is driven off, leaving the **anhydrous** salt behind. Hydrated barium chloride is a commonly used example because the barium chloride itself doesn't decompose even on quite strong heating.

If you heated barium chloride crystals in a crucible you might obtain these results:

Mass of crucible = 30.00 g

Mass of crucible + barium chloride crystals, $BaCl_2 \cdot nH_2O$ = 32.44 g

Mass of crucible + anhydrous barium chloride, $BaCl_2$ = 32.08 g

To find *n*, you need to find the ratio of the number of moles of $BaCl_2$ to the number of moles of water. It's just another empirical formula calculation (A_r of H = 1, A_r of O = 16, A_r of Cl = 35.5, A_r of Ba = 137).

Mass of $BaCl_2$ = 32.08 – 30.00 = 2.08 g

Mass of water = 32.44 – 32.08 = 0.36 g

Table 5.2 Finding the *n* in $BaCl_2 \cdot nH_2O$

	$BaCl_2$	H_2O
masses/g	2.08	0.36
divide by M_r to find the number of moles	2.08/208	0.36/18
number of moles	0.0100	0.020
ratio of moles	1	2
empirical formula	$BaCl_2 \cdot 2H_2O$	

CALCULATIONS USING MOLES, CHEMICAL EQUATIONS AND MASSES OF SUBSTANCES

INTERPRETING SYMBOLS IN EQUATIONS IN TERMS OF MOLES

KEY POINT

The big numbers in front of the formulae tell us the number of moles of each substance that react. Although there does not appear to be a big number in front of the CH_4 and CO_2, there should actually be a '1' in front of each but we just don't write it.

If we write a chemical equation:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

we can say that 1 molecule of CH_4 combines with 2 molecules of O_2 to form 1 molecule of CO_2 and 2 molecules of water. When doing calculations, however, it is often more useful to think about this in terms of moles and take symbols as meaning moles of each substance. So here we can say:

1 mol CH_4 reacts with 2 mol O_2 to form 1 mol CO_2 and 2 mol H_2O

We have not changed the meaning of the equation as the ratios are still the same.

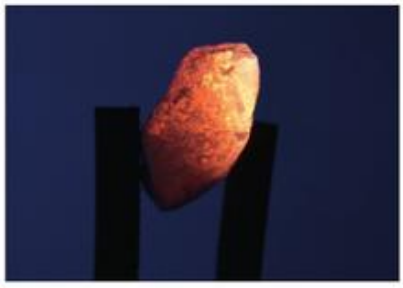
KEY POINT

Remember that 2 mol of O_2 contains twice as many molecules as 1 mol of CH_4.

We can therefore tell from this equation that 16 g of CH_4 (1 mol) reacts exactly with 64 g of O_2 (2 mol) to form 44 g of CO_2 (1 mol) and 36 g of H_2O (2 mol).

CALCULATIONS INVOLVING ONLY MASSES

Typical calculations will give you a mass of starting material and ask you to calculate how much product you are likely to obtain. You will also find examples done in reverse, where you are told the mass of the product and are asked to find out how much of the starting material you would need. In almost all the cases you will see at International GCSE you will be given the equation for the reaction.



▲ Figure 5.16 A glowing piece of limestone. Limestone is impure calcium carbonate – it decomposes when heated to form calcium oxide (quicklime) and carbon dioxide. This is a thermal decomposition reaction.

A PROBLEM INVOLVING HEATING CALCIUM CARBONATE

When calcium carbonate, $CaCO_3$, is heated, calcium oxide is formed. Imagine you wanted to calculate the mass of calcium oxide produced by heating 25 g of calcium carbonate (A_r: C = 12, O = 16, Ca = 40).

$$CaCO_3 \rightarrow CaO + CO_2$$

THE CALCULATION

The method we will use has three steps:

Step 1: Calculate the number of moles using the mass that you have been given.

Step 2: Use the chemical equation to work out the number of moles of the substance you are interested in.

Step 3: Convert the number of moles to a mass.

KEY POINT

The M_r of $CaCO_3$ is 100 and the M_r of CaO is 56.

In this reaction we have 25 g of $CaCO_3$ and the number of moles can be calculated as:

$$\text{number of moles} = \frac{\text{mass}}{\text{mass of 1 mole}} = \frac{25}{100} = 0.25\,\text{mol}$$

The chemical equation for this reaction is

$$CaCO_3 \rightarrow CaO + CO_2$$

We can see from the equation that 1 mole of $CaCO_3$ will decompose to produce 1 mole of CaO. In other words, the equation is telling us that if we start with a certain number of moles of $CaCO_3$ we will obtain the same number of moles of CaO at the end.

Therefore, we can deduce that 0.25 mol $CaCO_3$ will decompose to produce 0.25 mol CaO. Now that we know the number of moles of CaO (M_r = 56) we can convert this to a mass:

$$\text{mass} = \text{number of moles} \times \text{mass of 1 mole}$$

$$\text{mass} = 0.25 \times 56 = 14\,\text{g}$$

Therefore, the reaction will produce 14 g of calcium oxide.

ALTERNATIVE METHOD

The calculation can also be done in a different way using ratios:

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

Interpret the equation in terms of moles:

1 mol $CaCO_3$ produces 1 mol CaO (and 1 mol CO_2)

Substitute masses where relevant:

100 g (1 mol) $CaCO_3$ produces 56 g (1 mol) CaO

Do a proportion calculation:

If 100 g of calcium carbonate gives 56 g of calcium oxide

1 g of calcium carbonate gives $\frac{56}{100}$ g of calcium oxide = 0.56 g

25 g of calcium carbonate gives 25 × 0.56 g of calcium oxide = 14 g of calcium oxide

HINT

You do not have to learn both methods, just learn whichever method you are happier with.
Your maths may be so good that you don't need to take all these steps to find the answer. If you can do it more quickly, that's fine. However, you must still show all your calculations.
When you have finished a chemistry calculation, the impression should nearly always be that there are a lot of words with a few numbers between them, not vice versa.

A PROBLEM ABOUT EXTRACTING IRON

The equation below is for a reaction that occurs in the extraction of iron:

$Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$

Calculate the mass of iron which can be formed from 1000 g of iron oxide.

We are given the mass of Fe_2O_3 in the question and so we can calculate the number of moles of Fe_2O_3:

$$\text{number of moles} = \frac{\text{mass}}{\text{mass of 1 mole}}$$

$$\text{number of moles} = \frac{1000}{160} = 6.25 \text{ mol}$$

KEY POINT

The A_r of Fe is 56.
The M_r of $Fe_2O_3 = 2 \times 56 + 3 \times 18 = 160$.

From the chemical equation we can see that 1 mol Fe_2O_3 produces 2 mol Fe. We know this because of the big numbers in front of the formulae.

So we know that, if we start with a certain number of moles of Fe_2O_3, we will obtain twice as many moles of Fe.

So 6.25 mol Fe_2O_3 produces 2 × 6.25 = 12.5 mol Fe:

mass = number of moles × mass of 1 mole

The mass of 12.5 mol Fe is 12.5 × 56 = 700 g.

HINT

To find the mass of Fe we multiply the number of moles of Fe by the mass of 1 mole of Fe (56 g). A common mistake is to multiply the number of moles by the mass of 2Fe (112 g), but we have already used the '2' when we worked out the number of moles of Fe.

ALTERNATIVE METHOD IN TERMS OF RATIOS

First interpret the equation in terms of moles:

1 mol Fe_2O_3 reacts with C to form 2 mol Fe (and CO_2).

We are only looking at how much iron is produced, so let's introduce just these masses.

160 g (1 mol) of Fe_2O_3 produces 2 × 56 g (2 mol) of Fe.

That is, 160 g of Fe_2O_3 produces 112 g of Fe.

From this we can calculate that 1 g of Fe_2O_3 will produce 112/160 g of Fe, and therefore 1000 g of Fe will produce 1000 × 112/160 = 700 g.

A PROBLEM INVOLVING THE EXTRACTION OF LEAD

Lead is extracted from galena, PbS. The ore is roasted in air to produce lead(II) oxide, PbO:

$$2PbS(s) + 3O_2(g) \rightarrow 2PbO(s) + 2SO_2(g)$$

The lead(II) oxide is then converted to lead by heating it with carbon in a blast furnace:

$$PbO(s) + C(s) \rightarrow Pb(l) + CO(g)$$

The molten lead is tapped from the bottom of the furnace.

Calculate the mass of lead that would be produced from 1 tonne of galena.

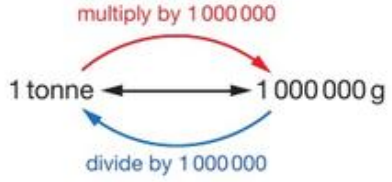
(A_r: O = 16, S = 32, Pb = 207)



▲ Figure 5.17 Molten lead tapped from the bottom of a furnace.

HINT

You should be told how many grams there are in a tonne if you need this information in the exam.



▲ Figure 5.18 How to convert between grams and tonnes

HINT

1 000 000 might be given to you as 10^6 or 1×10^6. These are both the same.

We have been given the mass of PbS so we can calculate the number of moles. However, we have to be careful because the mass is in tonnes but to calculate a number of moles we need to know the mass in grams, so we must convert tonnes to grams. The conversion for this is shown in Figure 5.18. Just write down this relationship: *1* tonne is equivalent to *1 000 000* g so, to change from tonnes to grams, the number has to *increase* (from 1 to 1 000 000), so we *multiply* by 1 000 000.

Therefore we have 1 000 000 g PbS.

$$\text{Number of moles} = \frac{\text{mass}}{\text{mass of 1 mole}}$$

$$\text{The number of moles of PbS} = \frac{1\,000\,000}{239} = 4184 \text{ mol.}$$

From the first equation:

$$2PbS(s) + 3O_2(g) \rightarrow 2PbO(s) + 2SO_2(g)$$

we can see that 2 mol PbS produce 2 mol PbO, therefore we can work out that 4184 mol PbS will produce 4184 mol PbO.

From the second equation:

$$PbO(s) + C(s) \rightarrow Pb(l) + CO(g)$$

we can see that 1 mol PbO produces 1 mol Pb. But we are starting this reaction with 4184 mol PbO and so we can work out that we will produce 4184 mol Pb.

The mass of 4184 mol Pb is $4184 \times 207 = 866\,000$ g.

We can convert this to tonnes by dividing by 1 000 000:

$$866\,000/1\,000\,000 = 0.866 \text{ tonne}$$

ALTERNATIVE METHOD

KEY POINT

We've doubled the second equation so that we can trace what happens to all the 2PbO from the first one.
This could also be simplified to 1 mol PbS produces 1 mol Pb. This would save you doing some arithmetic as you would not have to multiply everything by 2. However, in the end it does not make any difference to the answer, so you do not have to simplify it if you don't want to.

HINT

There are advantages and disadvantages of both methods. For the first method, we can use the same three steps in all the calculations we do below. However, a bit more maths will be involved: converting tonnes to grams and back again. The alternative method has fewer steps but we cannot use it in the same way when doing some of the calculations involving solutions in the next chapter.

$$2PbS(s) + 3O_2(g) \rightarrow 2PbO(s) + 2SO_2(g)$$

$$PbO(s) + C(s) \rightarrow Pb(l) + CO(g)$$

Interpret the equation in terms of moles and trace the lead through the equations:

2 mol PbS produces 2 mol PbO

If we double the second equation:

$$2PbO(s) + 2C(s) \rightarrow 2Pb(l) + 2CO(g)$$

we can see that 2 mol PbO produces 2 mol Pb.

In other words, every 2 mol of PbS produces 2 mol of Pb.

Substitute masses where relevant. In this case, the relevant masses are only the PbS and the Pb:

2 × 239 g PbS produces 2 × 207 g Pb

478 g PbS produces 414 g Pb

Now there seems to be a problem. The question is asking about tonnes and not grams. You could calculate how many grams there are in a tonne. However, it's much easier to think a bit, and realise that the ratio is always going to be the same, whatever the units, which means that:

478 tonnes PbS produces 414 tonnes Pb

Do the proportion calculation:

If 478 tonnes PbS produces 414 tonnes Pb

then 1 tonne PbS gives $\frac{417}{478}$ tonne Pb = 0.866 tonne

0.866 tonne of lead is produced from 1 tonne of galena.

CALCULATING PERCENTAGE YIELDS

WHAT IS A PERCENTAGE YIELD?

If you calculate how much product a reaction might produce, in real life you rarely obtain as much as you expected. If you expect to get 100 g, but only get 80 g, your percentage yield is 80%. The rest of it has been lost in some way. This could be due to spillages, or losses when you transfer a liquid from one container to another. Or it may be that there are all sorts of side reactions going on, so that some of your starting materials are changed into unwanted products. That happens a lot during reactions in organic chemistry.



▶ Figure 5.19 Experiments done by students to make and purify organic compounds rarely give a high percentage yield.

CALCULATING THE PERCENTAGE YIELD

HINT

The percentage yield should always come out as less than 100%. If it comes out as more than 100% either you have made a mistake in the calculation or the question will continue by asking about what might have gone wrong in the experiment to give a value greater than 100%. If you are making a salt (see Chapter 17), for instance, a possible reason you could get a percentage yield greater than 100% is that it was not completely dry.

Suppose you do a calculation and work out that you should get 12.5 g of copper sulfate crystals, but you only get 11.2 g when you actually do the experiment. 12.5 g is the *theoretical yield* and can be obtained by carrying out a moles calculation. 11.2 g is the *actual yield* – this is what is obtained in the actual experiment and will be given to you in the question.

$$\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

The percentage yield is $\frac{11.2}{12.5} \times 100 = 89.6\%$.

EXAMPLE 7

A student reacted 2.40 g of copper(II) oxide (CuO) with hot sulfuric acid. She made 5.21 g of copper(II) sulfate crystals ($CuSO_4 \cdot 5H_2O$).

The equations for the reactions are:

$$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$$

$$CuSO_4(aq) + 5H_2O(l) \rightarrow CuSO_4 \cdot 5H_2O(l)$$

First we need to calculate the theoretical yield. This is done by carrying out a moles calculation. We have enough information to calculate the number of moles of copper(II) oxide:

$$\text{number of moles of CuO} = \frac{2.40}{79.5} = 0.0302\,\text{mol}$$

From the first equation we can deduce that 0.0302 mol CuO will produce 0.0302 mol $CuSO_4$.

From the second equation we can deduce that 0.0302 mol $CuSO_4$ will produce 0.0302 mol $CuSO_4 \cdot 5H_2O$.

Now we need to calculate the mass of $CuSO_4 \cdot 5H_2O$ by multiplying the number of moles by the mass of 1 mole:

$$\text{mass} = 0.0302 \times 249.5 = 7.53\,\text{g}$$

This is the theoretical yield of copper(II) sulfate crystals.

The actual yield is what was obtained in the experiment. This was 5.21 g.

The percentage yield = (actual yield/theoretical yield) × 100

The percentage yield is 5.21/7.53 × 100 = 69.2%.

CALCULATIONS IN WHICH YOU HAVE TO CALCULATE WHICH SUBSTANCE IS IN EXCESS

When you make something in the lab, you rarely mix things together in exactly the right proportions. Usually, something is in excess, and you remove the excess (by filtering, for example) when the reaction is complete. If something is in excess there is more than enough of it to react with the other substance and so we don't need to worry about it.

In all the examples we have done above, we have only ever been given enough information to calculate the number of moles of one thing at the beginning of the problem. We have used this number of moles throughout the calculations. But what would happen if we were given the following question?

Magnesium reacts with hydrochloric acid according to the following equation:

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

0.2 mol Mg is reacted with 0.2 mol HCl. Calculate the mass of hydrogen gas produced.

We might be tempted to say that 0.2 mol Mg produces 0.2 mol H_2(g) and the mass of $H_2 = 0.2 \times 2 = 0.4$ g. Unfortunately this is not the correct answer!

We can understand why it is not correct by looking at the chemical equation. 1 mol Mg reacts with 2 mol HCl and therefore 0.2 mol Mg would react with $2 \times 0.2 = 0.4$ mol HCl.

But we only used 0.2 mol HCl and so we did not have enough to react with all the Mg. Because all the Mg did not react we cannot use the number of moles of Mg to work out how much hydrogen is produced.

If we try starting with the number of moles of HCl, we can deduce from the chemical equation that 2 mol HCl reacts with 1 mol Mg, therefore 0.2 mol HCl reacts with 0.1 mol Mg. We added 0.2 mol Mg and so there is more than enough to react with all the HCl – the Mg is in **excess**.

KEY POINT

You always have to use the chemical equation to determine which substance is in excess. If we had been given 0.2 mol Mg and 0.3 mol HCl in this reaction it would look like the HCl is in excess because there is more of it. But if you look at the chemical equation you can see that 0.3 mol HCl only reacts with 0.15 mol Mg so there is still not enough HCl to react with all the Mg, so the Mg is still in excess.

Because we know that all the HCl reacts we can use the number of moles of HCl to work out how much H_2 is produced.

2 mol HCl produce 1 mol H_2: the number of moles of H_2 is half the number of moles of HCl.

Therefore 0.2 mol HCl produce 0.1 mol H_2.

The mass of $H_2 = 0.1 \times 2 = 0.2$ g.

This is the correct answer!

To deduce which substance is in excess we try both numbers of moles in the chemical equation and compare the numbers.

KEY POINT

A reagent is just something that reacts.

EXAMPLE 8

Copper reacts with concentrated nitric acid according to the equation:

$$Cu(s) + 4HNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2H_2O(l) + 2NO_2(g)$$

3.2 g of copper is reacted with 0.40 mol concentrated nitric acid. Work out which reagent is in excess.

We need to convert the mass of Cu to a number of moles:

$$\text{number of moles of Cu} = \frac{3.2}{63.5} = 0.050 \text{ mol}$$

Looking at the chemical equation we can see that 1 mol Cu reacts with 4 mol HNO_3, so we can calculate that:

0.050 mol Cu reacts with $4 \times 0.050 = 0.20$ mol HNO_3

We have more than 0.20 mol HNO_3 therefore HNO_3 is in excess.

If the question went further and asked us to calculate the amount in moles of NO_2 produced using these quantities we would have to use the number of moles of Cu because not all the HNO_3 is used up. We would obtain $2 \times 0.050 = 0.10$ mol NO_2.

CHAPTER QUESTIONS

SKILLS PROBLEM SOLVING

8

1 Balance the following equations:

a $Fe + HCl \rightarrow FeCl_2 + H_2$

b $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

c $Ca + H_2O \rightarrow Ca(OH)_2 + H_2$

d $Al + Cr_2O_3 \rightarrow Al_2O_3 + Cr$

e $Fe_2O_3 + CO \rightarrow Fe + CO_2$

f $NaHCO_3 + H_2SO_4 \rightarrow Na_2SO_4 + CO_2 + H_2O$

g $C_8H_{18} + O_2 \rightarrow CO_2 + H_2O$

h $Fe_3O_4 + H_2 \rightarrow Fe + H_2O$

i $Pb + AgNO_3 \rightarrow Pb(NO_3)_2 + Ag$

j $AgNO_3 + MgCl_2 \rightarrow Mg(NO_3)_2 + AgCl$

k $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

2 Calculate the relative formula masses of the following compounds:

a CO_2

b CH_3CO_2H

c Na_2SO_4

d $(NH_4)_2SO_4$

e $Na_2CO_3 \cdot 10H_2O$

f $Cr_2(SO_4)_3$

(A_r: H = 1, C = 12, N = 14, O = 16, Na = 23, S = 32, Ca = 40, Cr = 52, Fe = 56)

9

3 Calculate the percentage of nitrogen in each of the following substances (all used as nitrogen fertilisers):

a potassium nitrate, KNO_3

b ammonium nitrate, NH_4NO_3

c ammonium sulfate, $(NH_4)_2SO_4$

(A_r: H = 1, N = 14, O = 16, S = 32, K = 39)

4 Calculate the mass of the following:

a 1 mol of HCN

b 1 mol of lead(II) nitrate, $Pb(NO_3)_2$

c 4.30 mol of methane, CH_4

d 0.70 mol of Na_2O

e 0.015 mol of $NaNO_3$

f 0.24 mol of sodium carbonate crystals, $Na_2CO_3 \cdot 10H_2O$

(A_r: H = 1, C = 12, N = 14, O = 16, Na = 23, Pb = 207)

SKILLS PROBLEM SOLVING



5 Calculate the number of moles represented by each of the following:

a 20 g of magnesium oxide, MgO

b 3.20 g of iron(III) oxide, Fe_2O_3

c 2 kg of copper(II) oxide, CuO

d 50 g of copper(II) sulfate crystals, $CuSO_4 \cdot 5H_2O$

e 1 tonne of iron, Fe (1 tonne is 1 000 000 g)

f 0.032 g of sulfur dioxide, SO_2

(A_r: H = 1, O = 16, S = 32, Mg = 24, Fe = 56, Cu = 63.5)

6 The following calculations use A_r: H = 1, O = 16, Na = 23, Cl = 35.5, Ca = 40, Cu = 63.5.

a Calculate the mass of 4 mol of sodium chloride, NaCl.

b Calculate how many moles is 37 g of calcium hydroxide, $Ca(OH)_2$.

c Calculate how many moles is 1 kg (1000 g) of calcium, Ca.

d Calculate the mass of 0.125 mol of copper(II) oxide, CuO.

e 0.1 mol of a substance has a mass of 4 g. Calculate the mass of 1 mole.

f 0.004 mol of a substance has a mass of 1 g. Calculate the relative formula mass of the compound.

7 Determine the empirical formulae of the compounds which contain:

a 9.39 g P, 0.61 g H

b 5.85 g K, 2.10 g N, 4.80 g O

c 3.22 g Na, 4.48 g S, 3.36 g O

d 22.0% C, 4.6% H, 73.4% Br (by mass)

(A_r: H = 1, C = 12, N = 14, O = 16, Na = 23, S = 32, K = 39, Br = 80)

8 1.24 g of phosphorus was burned completely in oxygen to give 2.84 g of phosphorus oxide. Find:

a the empirical formula of the oxide

b the molecular formula of the oxide given that 1 mole of the oxide has a mass of 284 g.

(A_r: O = 16, P = 31)

9 An organic compound contained C 66.7%, H 11.1%, O 22.2% by mass. Its relative formula mass was 72. Find:

a the empirical formula of the compound

b the molecular formula of the compound.

(A_r: H = 1, C = 12, O = 16)

10 In an experiment to find the number of molecules of water of crystallisation in sodium sulfate crystals, $Na_2SO_4 \cdot nH_2O$, 3.22 g of sodium sulfate crystals were heated gently. When all the water of crystallisation had been driven off, 1.42 g of anhydrous sodium sulfate was left. Find the value of n in the formula. (A_r: H = 1, O = 16, Na = 23, S = 32)

SKILLS PROBLEM SOLVING

11 Gypsum is hydrated calcium sulfate, $CaSO_4 \cdot nH_2O$. A sample of gypsum was heated in a crucible until all the water of crystallisation had been driven off. The following results were obtained:

Mass of crucible = 37.34 g

Mass of crucible + gypsum, $CaSO_4 \cdot nH_2O$ = 45.94 g

Mass of crucible + anhydrous calcium sulfate, $CaSO_4$ = 44.14 g

Calculate the value of n in the formula $CaSO_4 \cdot nH_2O$.

(A_r: H = 1, O = 16, S = 32, Ca = 40)

12 **a** Calculate the amount in moles of SO_3 formed when 0.36 mol SO_2 reacts with excess O_2.

$2SO_2 + O_2 \rightarrow 2SO_3$

b Calculate the amount in moles of HCl that reacts with 0.4 mol $CaCO_3$.

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

c Calculate the amount in moles of H_2S formed when 0.4 mol of HCl reacts with excess Sb_2S_3.

$Sb_2S_3 + 6HCl \rightarrow 2SbCl_3 + 3H_2S$

d Calculate the amount in moles of iron formed when 0.9 mol carbon monoxide reacts with excess iron(III) oxide.

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

e Calculate the amount in moles of hydrogen that would be required to make 0.8 mol NH_3.

$N_2 + 3H_2 \rightarrow 2NH_3$

13 The reaction between iron and bromine is

$2Fe + 3Br_2 \rightarrow 2FeBr_3$

10 g of iron is reacted with bromine.

a Calculate the amount in moles of iron that reacted.

b Calculate the amount in moles of bromine that the iron reacted with.

c Calculate the amount in moles of $FeBr_3$ that will be formed.

d Calculate the mass of $FeBr_3$ formed.

(A_r: Fe = 56, Br = 80)

14 Titanium is manufactured by heating titanium(IV) chloride with sodium.

$TiCl_4(g) + 4Na(l) \rightarrow Ti(s) + 4NaCl(s)$

1.0 g of $TiCl_4$ is reacted with excess sodium

(A_r: Na = 23, Ti = 48, Cl = 35.5)

a Calculate the amount in moles of $TiCl_4$ reacted.

b Calculate the amount in moles of Ti formed.

c Calculate the mass of Ti formed.

d Calculate the mass of NaCl formed.

e Calculate the mass of Ti formed when 1 tonne of $TiCl_4$ reacts with excess sodium (1 tonne is 1 000 000 g).

SKILLS PROBLEM SOLVING

15 2.67 g of aluminium chloride was dissolved in water and an excess of silver nitrate solution was added to give a precipitate of silver chloride (AgCl).

$AlCl_3(aq) + 3AgNO_3(aq) \rightarrow Al(NO_3)_3(aq) + 3AgCl(s)$

Calculate the mass of silver chloride that would be formed.

(A_r: Al = 27, Cl = 35.5, Ag = 108)

16 Chromium is manufactured by heating a mixture of chromium(III) oxide with aluminium powder.

$Cr_2O_3(s) + 2Al(s) \rightarrow 2Cr(s) + Al_2O_3(s)$

a Calculate the mass of aluminium needed to react with 50 g of Cr_2O_3.

b Calculate the mass of chromium produced from 50 g of Cr_2O_3.

c Calculate the mass of chromium produced from 5 kg of Cr_2O_3.

d Calculate the mass of chromium produced from 5 tonnes of Cr_2O_3.

(1 tonne is 1 000 000 g)

(A_r: Cr = 52, O = 16, Al = 27)

17 Copper(II) sulfate crystals, $CuSO_4 \cdot 5H_2O$, can be made by heating copper(II) oxide with dilute sulfuric acid and then crystallising the solution formed.

a Calculate the maximum mass of crystals that could be made from 4.00 g of copper(II) oxide using an excess of sulfuric acid.

$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$

$CuSO_4(aq) + 5H_2O(l) \rightarrow CuSO_4 \cdot 5H_2O(s)$

b If the actual mass of copper(II) sulfate collected at the end of the experiment was 11.25 g, calculate the percentage yield.

(A_r: H = 1, O = 16, S = 32, Cu = 63.5)

18 Ethanol reacts with ethanoic acid to form ethyl ethanoate and water:

$C_2H_5OH(l) + CH_3COOH(l) \rightarrow CH_3COOC_2H_5(l) + H_2O(l)$

ethanol ethanoic acid ethyl ethanoate water

a Calculate the mass of ethyl ethanoate produced when 20.0 g of ethanol reacts with excess ethanoic acid.

b The actual yield in this experiment is 30.0 g of ethyl ethanoate. Calculate the percentage yield.

(A_r: C = 12, H = 1, O = 16)

SKILLS REASONING

19 Determine which reactant is in excess in each case.

a 0.5 mol of Na_2CO_3 is reacted with 0.5 mol hydrochloric acid:

$Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$

b 0.01 mol C_3H_8 is reacted with 0.02 mol O_2:

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

c 28 g CO is reacted with 0.6 mol Fe_2O_3:

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

d 16 g O_2 is reacted with 16 g SO_2:

$2SO_2 + O_2 \rightarrow 2SO_3$

(A_r : C = 12, O = 16, S = 32)

20 1.0 g of $CaCO_3$ is reacted with 0.015 mol of HCl:

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

a Determine which reagent is present in excess.

b Calculate the mass of CO_2 produced.

(A_r: Ca = 40, C = 12, O = 16)

SKILLS PROBLEM SOLVING